Data & AI Builders Conference

Building modern data foundations for intelligent solutions together

Multi-Cloud data platform integration with OneLake Shortcut and Caching

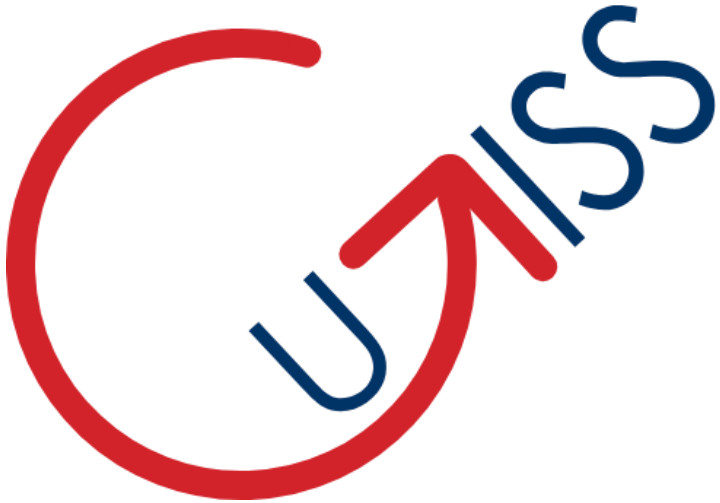
Riccardo Perico
Analytics & AI Architect @ Lucient |
Data Platform MVP



Sponsors



Collaborators





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Analytics & AI Architect @ Lucient
+15 years in Microsoft “Data Realm”
Giving my little contribution to the community

PASS
PBIUG ITALY
POWER PLATFORM BOOTCAMP
Power Platform World Tour
DATA SATURDAYS
DevMarche
Lodestar

“OneLake is the least “shiny” Microsoft Fabric component but it’s most the precious”

Riccardo Perico ☺



1 oz Gold = 4300 USD
1 oz Rhodium = 8800 USD
(Aug ’26)

1 oz Californium = 800 Million UDS



What is it OneLake?

Lodestar

OneLake for all data

"The OneDrive for data"



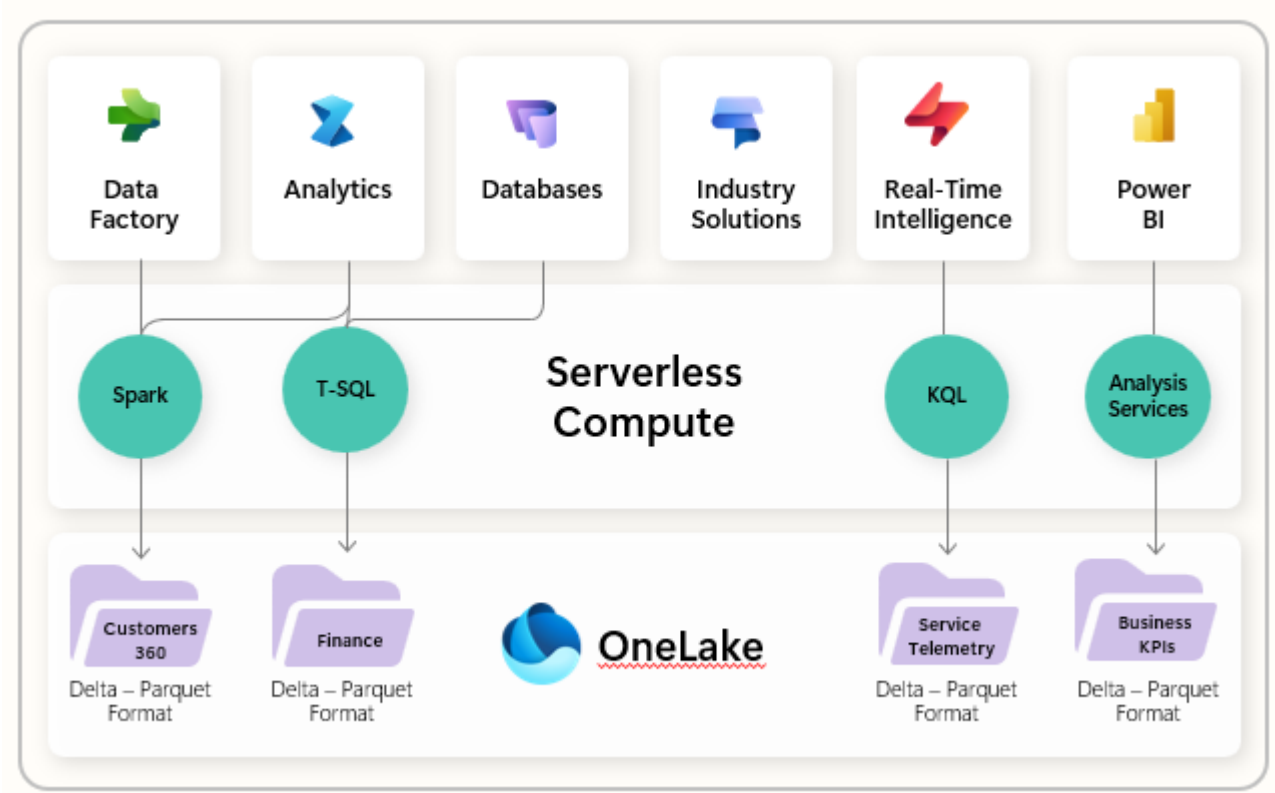
OneDrive

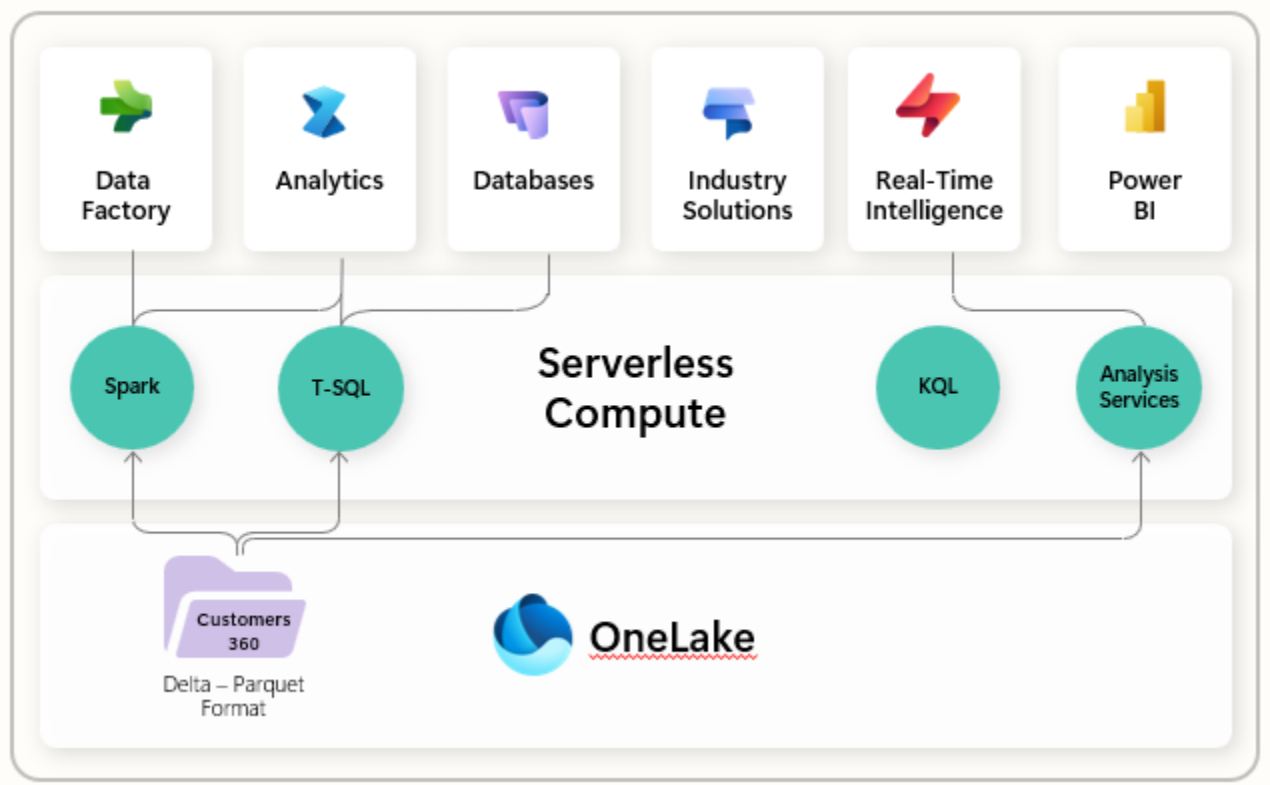


OneLake

OneLake provides a data lake as a service without you needing to build it




Data Factory
Analytics
Databases
Industry Solutions
Real-Time Intelligence
Power BI
Spark
T-SQL
Serverless Compute
KQL
Analysis Services
Customers 360
Delta – Parquet Format
Finance
Delta – Parquet Format
OneLake
Service Telemetry
Delta – Parquet Format
Business KPIs
Delta – Parquet Format


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OneLake



ICEBERG



Lodestar



Doesn’t seem so precious?

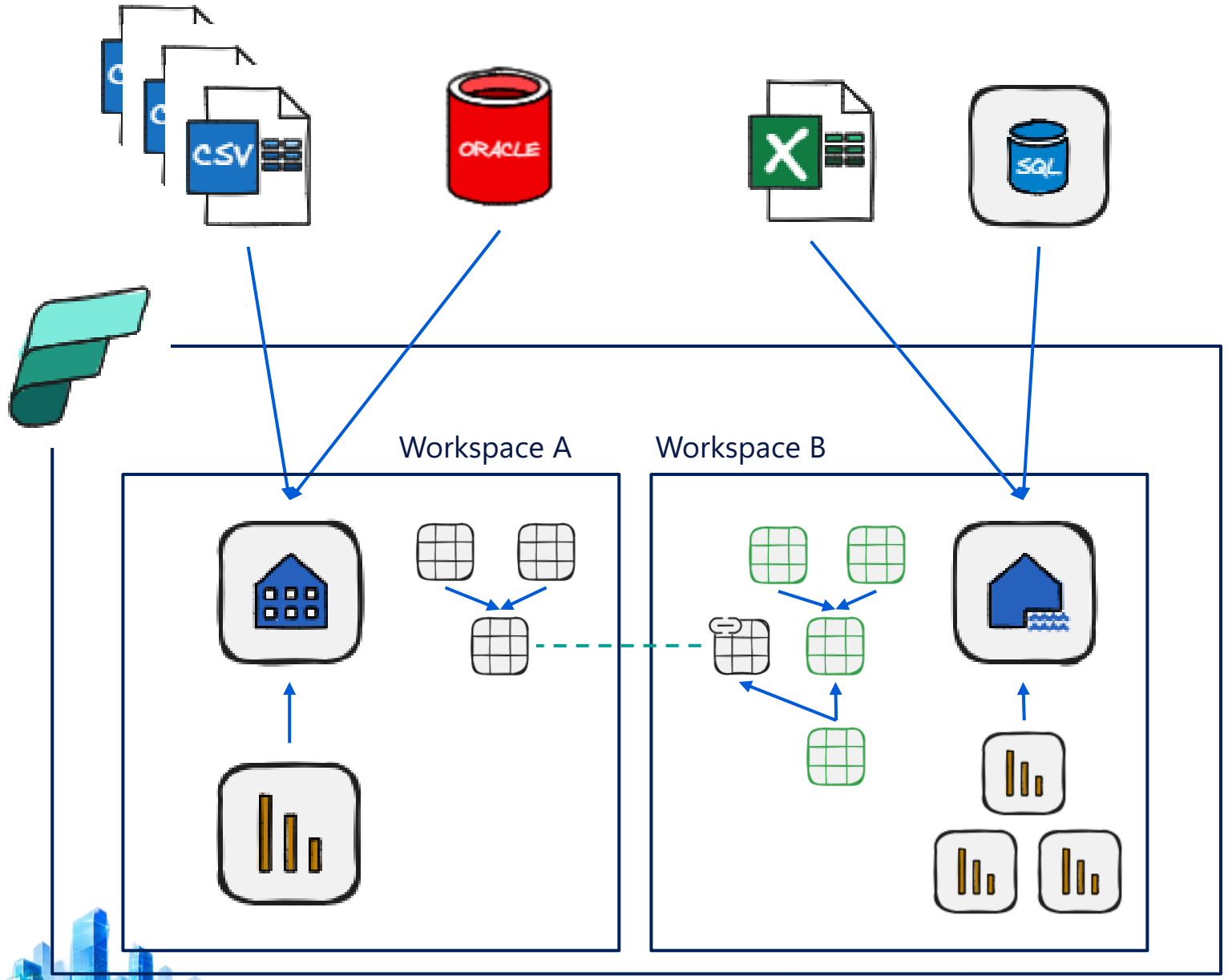
Lodestar


NOW (TODAY)
NEW BUILDING, HIDDEN RISK
BEAUTIFUL NEW SKYSCRAPER
BAD GROUND EVEN TODAY!
VERY WEAK SOIL
HIGH WATER TABLE
LOOSE & COMPRESSIBLE SOIL
POSSIBLE VOIDS & SETTLEMENT
LOOKS PERFECT TODAY... BUT THE GROUND IS UNSTABLE!
THEN (FUTURE)
TIME PASSES, PROBLEMS APPEAR
BUILDING STARTS TO TILT
CRACKS DEVELOP
DOORS & WINDOWS DON'T CLOSE PROPERLY
UNEQUAL SETTLEMENT DAMAGES THE STRUCTURE
OVER TIME, THE BAD GROUND CAUSES SERIOUS PROBLEMS AND HIGH COSTS!



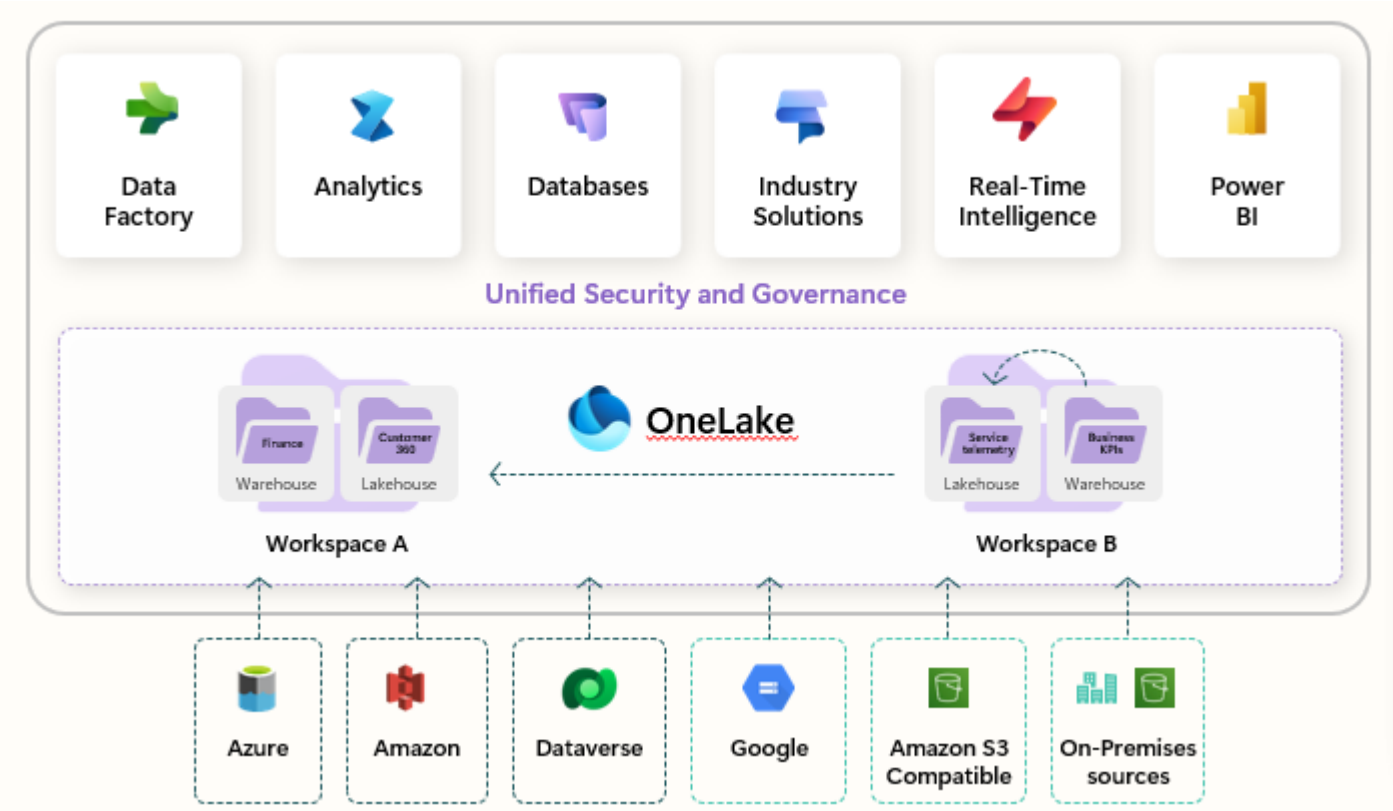
OneLake Shortcuts

Lodestar



Benefits:

1. Space saving
2. Time saving
3. CUs saving
4. Don't need to reinvent the wheel
5. Data ownership delegated
6. Fix once and for all
7. Add feature once and for all


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Lakehouse
Warehouse
Workspace A
Workspace B
Azure
Amazon
Dataverse
Google
Amazon S3 Compatible
On-Premises sources

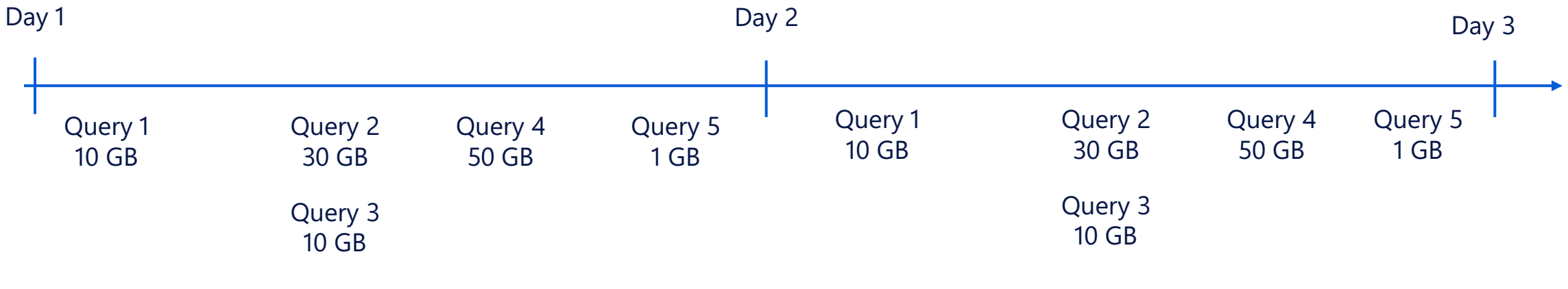
Lodestar

Networking is a factor

- Fabric to Shortcut source path could be long (Italy North -> West US) and **performance** may be impacted
- **Cloud providers charge you** for internet egress



Example on 1 Shortcut reading from AWS*



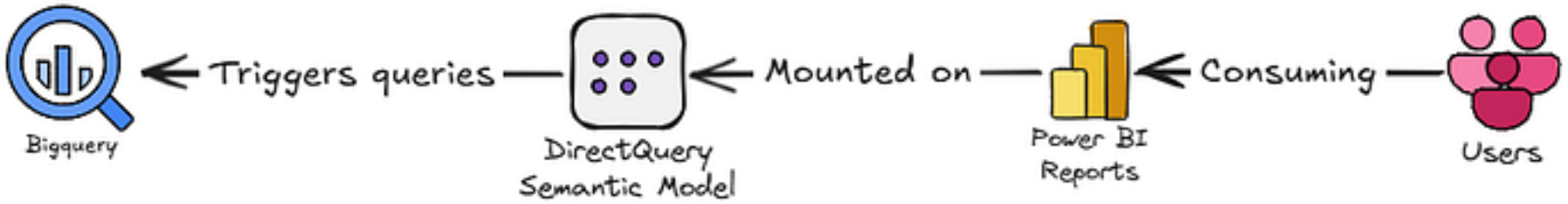
Data Read	10 GB	50 GB	100 GB	101 GB	111 GB	151 GB	201 GB	202 GB
Query Cost	0.00 USD	0.00 USD	0.00 USD	0.09 USD	0.90 USD	3.60 USD	4.50 USD	0.09 USD
Total Bill	0.00 USD	0.00 USD	0.00 USD	0.09 USD	0.99 USD	4.59 USD	9.09 USD	9.18 USD

Day 2 costed 100X Day 1

*prices may change

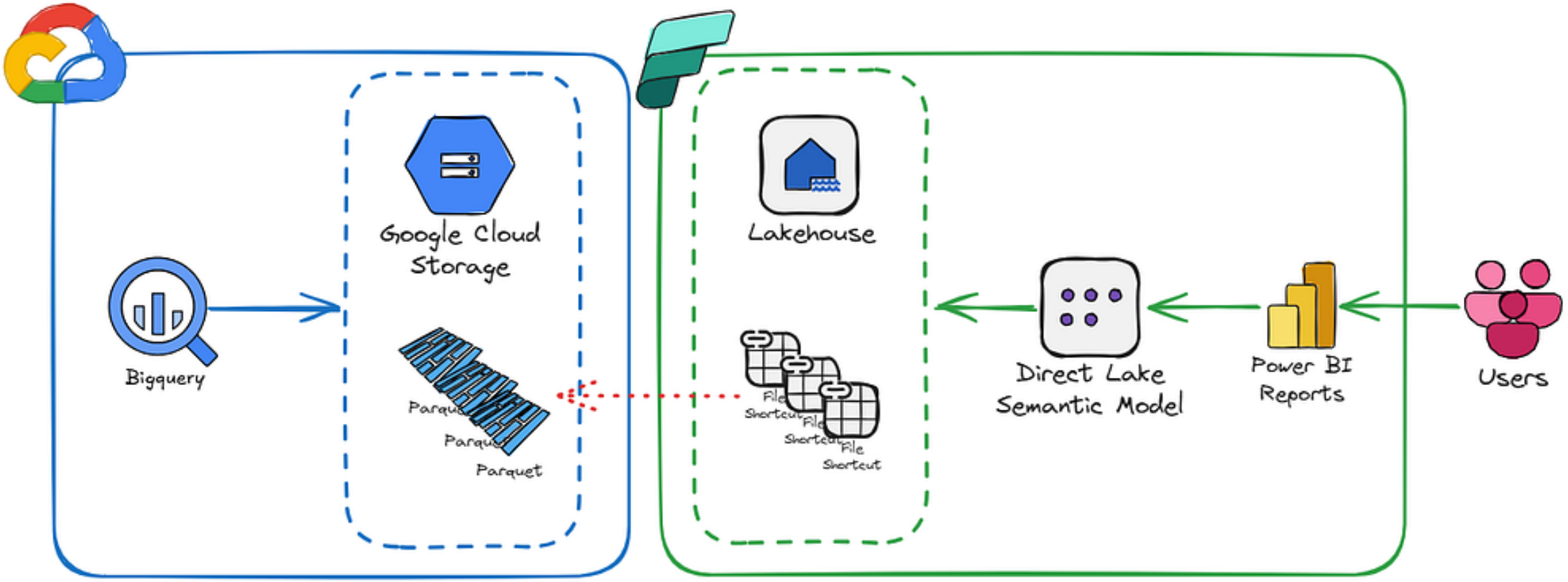
Ladestar

Scenario from the real world



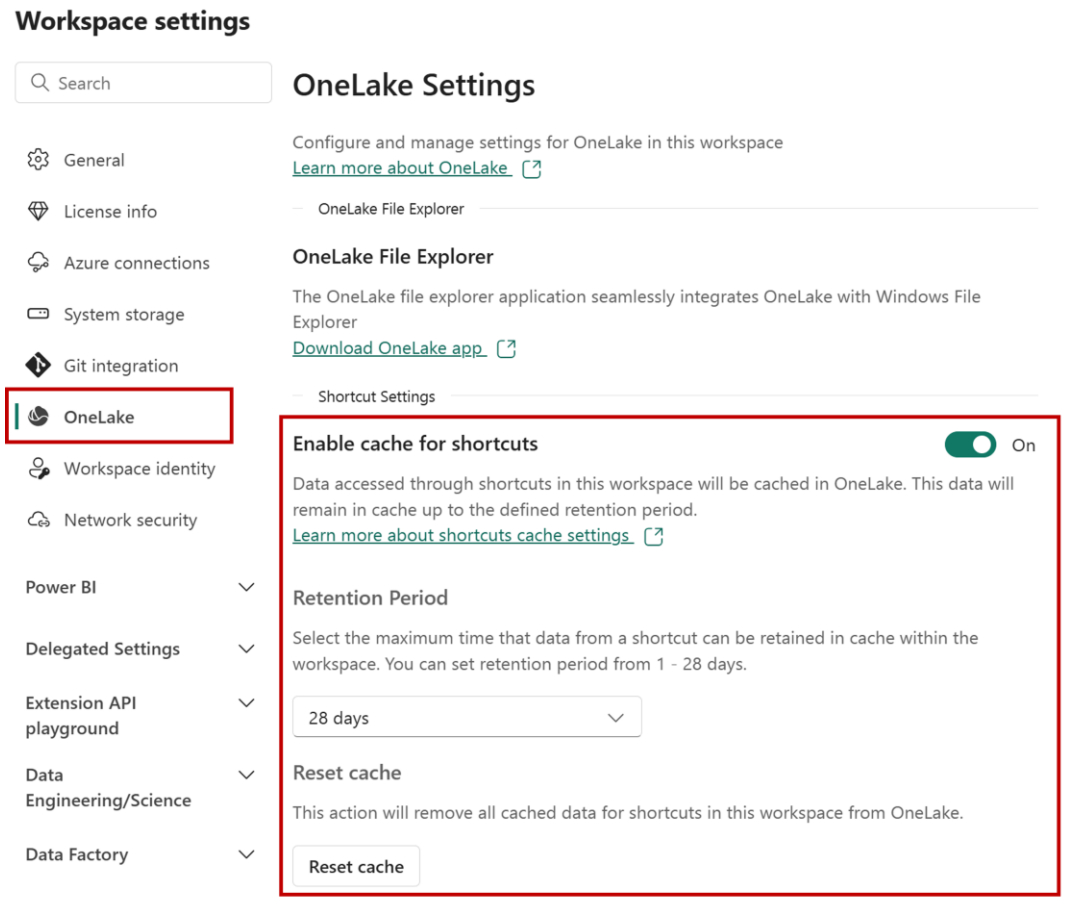
Big query DWH is loaded once per week

We proposed a rebuild



Problem: We're hitting GCS for any query of the user

OneLake caching to reduce Egress



Lodestar

Behind the scenes



- You set the retention between 1-28 days
- Supported:
  - GCS
  - S3
  - S3 Compatible (even on-prem)
- Up to 1GB/file
- First read from source subsequent from Fabric cache unless
  - Retention reached
  - Source has more recent file





Demo

Lodestar

SELECT *
FROM
myLH.dbo.mySC




```
"eventTime": "2026-08-19T15:29:44Z",
"eventSource": "s3.amazonaws.com",
"eventName": "HeadObject",
"awsRegion": "eu-north-1",
"sourceIPAddress": "20.38.86.39",
"userAgent": "[aws-sdk-dotnet-coreclr/3.7.304
"errorCode": "NoSuchKey",
"errorMessage": "The specified key does not e
"requestParameters": {
    "bucketName": "r1k91s3",
    "Host": "r1k91s3.s3.eu-north-1.amazonaws.
    "key": "titanic.parquet/"
},
"responseElements": null,
"additionalEventData": {
    "SignatureVersion": "SigV4",
    "CipherSuite": "TLS_AES_128_GCM_SHA256",
    "bytesTransferredIn": 0,
    "startTime": 1787153384944,
    "AuthenticationMethod": "AuthHeader",
    "endTime": 1787153384951,
    "x-amz-id-2": "sF0ZDz5N54sCMx6wvz+NGxtzxC
    "httpStatusCode": 404,
    "bytesTransferredOut": 287
},
```

```
"eventTime": "2026-08-19T15:29:45Z",
"eventSource": "s3.amazonaws.com",
"eventName": "GetObject",
"awsRegion": "eu-north-1",
"sourceIPAddress": "4.207.16.40",
"userAgent": "[aws-sdk-dotnet-coreclr/3.7.30
"requestParameters": {
    "bucketName": "r1k91s3",
    "Host": "r1k91s3.s3.eu-north-1.amazonaws
    "key": "titanic.parquet"
},
"responseElements": null,
"additionalEventData": {
    "SignatureVersion": "SigV4",
    "CipherSuite": "TLS_AES_128_GCM_SHA256",
    "objectSize": 40011,
    "bytesTransferredIn": 0,
    "startTime": 1787153385941,
    "AuthenticationMethod": "AuthHeader",
    "endTime": 1787153385965,
    "x-amz-id-2": "k0nn3T7fDJvNPHrU9i8+25Ism
    "turnAroundTime": 23,
    "httpStatusCode": 206,
    "bytesTransferredOut": 40011
},
```

SELECT *
FROM
myLH.dbo.mySC




```
"eventTime": "2026-08-19T15:45:07Z",
"eventSource": "s3.amazonaws.com",
"eventName": "HeadObject",
"awsRegion": "eu-north-1",
"sourceIPAddress": "20.38.86.97",
"userAgent": "[aws-sdk-dotnet-coreclr/3.7.304.5 ua/2.0 os/windows#10.0.20348.0 md/ARCH#X64 lang/.NET_Core#8.0.28 md/aws-sdk-dotnet-core#3.7.402.13 cfg/retry-mode#legacy md/ClientAsync
"errorCode": "NoSuchKey",
"errorMessage": "The specified key does not exist.",
"requestParameters": {
    "bucketName": "r1k91s3",
    "Host": "r1k91s3.s3.eu-north-1.amazonaws.com",
    "key": "titanic.parquet/"
},
"responseElements": null,
"additionalEventData": {
    "SignatureVersion": "SigV4",
    "CipherSuite": "TLS_AES_128_GCM_SHA256",
    "bytesTransferredIn": 0,
    "startTime": 1787154307261,
    "AuthenticationMethod": "AuthHeader",
    "endTime": 1787154307269,
    "x-amz-id-2": "pYgxxtogKYFhonXxDumovued6apOlTMUvOZCZ0+3SwK/PfOYxGdXhiEWP0+OF2Rg/m6qIXJ0gOFoAi1G/+lZeIqEiRx6iETO",
    "httpStatusCode": 404,
    "bytesTransferredOut": 307
```

```
SELECT *
FROM
myLH.dbo.mySC
```



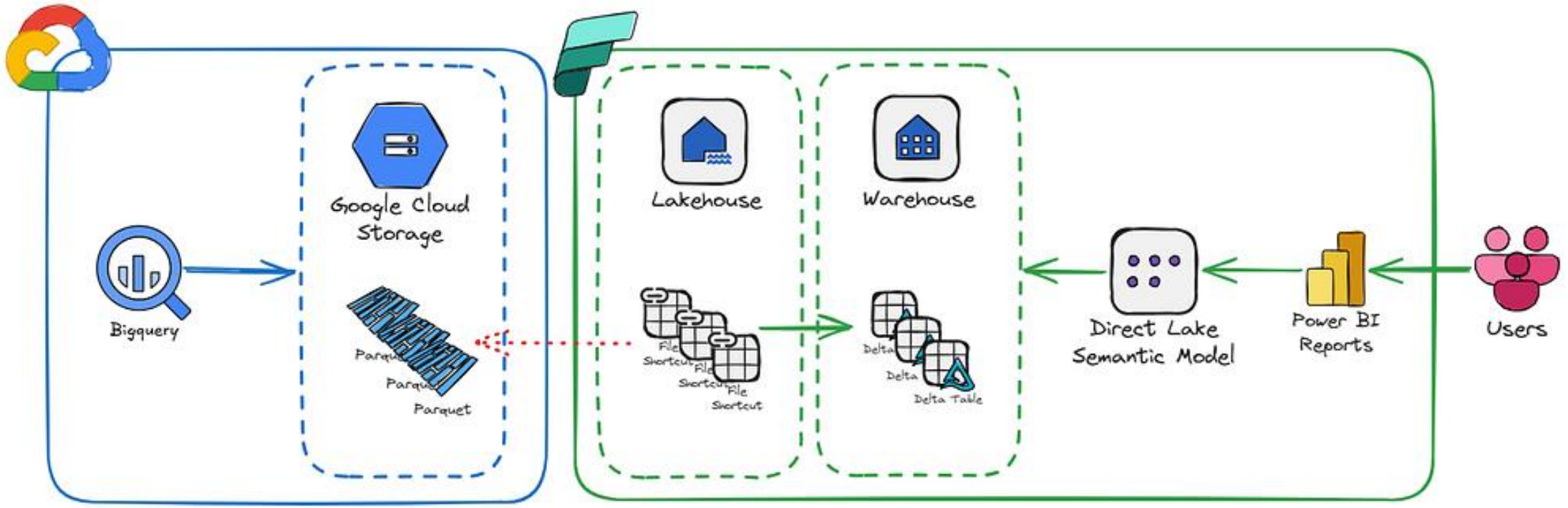

```
},
"eventTime": "2026-08-20T09:17:37Z",
"eventSource": "s3.amazonaws.com",
"eventName": "GetObject",
"awsRegion": "eu-north-1",
"sourceIPAddress": "20.38.85.21",
"userAgent": "[aws-sdk-dotnet-coreclr/3.7.304.5 ua
"requestParameters": {
    "bucketName": "r1k91s3",
    "Host": "r1k91s3.s3.eu-north-1.amazonaws.com",
    "key": "part-0.parquet"
},
"responseElements": null,
"additionalEventData": {
    "SignatureVersion": "SigV4",
    "CipherSuite": "TLS_AES_128_GCM_SHA256",
    "objectSize": 14759664,
    "bytesTransferredIn": 0,
    "startTime": 1787217457929,
    "AuthenticationMethod": "AuthHeader",
    "endTime": 1787217457949,
    "x-amz-id-2": "LQCNij9LJtmfUZVBOR9ddTm9GxSVF4
    "turnAroundTime": 19,
    "httpStatusCode": 206,
    "bytesTransferredOut": 65536
},
```

```
SELECT *
FROM
myLH.dbo.mySC
```




```
        "cipherSuite": "TLS_AES_128_GCM_SHA256",
        "clientProvidedHostHeader": "r1k91s3.s3.eu-north-1.amazonaws.com"
    }
},
{
    "eventVersion": "1.11",
    "userIdentity": {
        "type": "IAMUser",
        "principalId": "AIDA6HQ3U2GSB54XNNEIJ",
        "arn": "arn:aws:iam::978236789156:user/r1k91",
        "accountId": "978236789156",
        "accessKeyId": "AKIA6HQ3U2GSFXVGUU42",
        "userName": "r1k91"
    },
    "eventTime": "2026-08-19T16:44:29Z",
    "eventSource": "s3.amazonaws.com",
    "eventName": "HeadObject",
    "awsRegion": "eu-north-1",
    "sourceIPAddress": "20.38.85.19",
    "userAgent": "[aws-sdk-dotnet-coreclr/3.7.304.5 ua/2.0 os/windows#10.0.20348.0 md/ARCH#X64 lang/.NET_Core#8.0.28 md/aws-sdk-dotnet-core#3.7.402.13 cfg/retry-mode#legacy md/ClientAsync
    "errorCode": "NoSuchKey",
    "errorMessage": "The specified key does not exist.",
    "requestParameters": {
        "bucketName": "r1k91s3",
        "Host": "r1k91s3.s3.eu-north-1.amazonaws.com",
        "key": "part-0.parquet/"
    },
    "responseElements": null,
    "additionalEventData": {
        "SignatureVersion": "SigV4",
        "CipherSuite": "TLS_AES_128_GCM_SHA256",
        "bytesTransferredIn": 0,
        "startTime": 1787157869758,
        "AuthenticationMethod": "AuthHeader",
        "endTime": 1787157869766,
        "x-amz-id-2": "UZV12JwteDVBBQ/LKhlSgpRtiMjUsC75bu2pjKCTIlbAuZTA+scCXzO9DaQBfvxvC1bnKrnc1u37CirH24HTLc1jzZQdcvuz",
        "httpStatusCode": 404
```

The final layout



Savings on GC: 92%

Caveats

- Cache observability → you don’t know if a query hit the cache
- Cache visibility → you don’t know if the cache is ready
- Cache scope → at the workspace level



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THANK YOU !!

